

ARMAAN GUPTA

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🌐 armaan.world

Education

University of Waterloo

Bachelor of Computer Science + Minor in Artificial Intelligence (3.7+ GPA)

Sept. 2024 – Apr. 2029

Waterloo, ON

Technical Skills

Languages: Java, Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS, Swift, PHP, Ruby on Rails

Frameworks, Tools, & Libraries: React, Node.js, Git, AWS, TensorFlow, PyTorch, Firebase, Angular, Docker Next.js, npm, Flask, pandas, NumPy, Matplotlib, MongoDB, YOLO, OpenCV, Jupyter Notebooks, Google Colab, Tailwind, FastAPI, REST API, Vite, Three.js, Figma, Linux, Bash, GCP

Experience

Environment Canada (ECCC)

Software Engineer

May 2025 – Present

Toronto, ON

- Developed a climate reanalysis model pipeline by integrating **ML-based** interpolation techniques using **scikit-learn** and **PyTorch**, improving **spatial resolution** of missing environmental data by **35%**.
- Built an interactive web tool to explore **40+ years** of climate data across **50+** stations and **16 datasets** using Python, pandas, and D3.js, giving Canadians access to novel reanalysis datasets.
- Automated deployments and **version control** via **git**, **Docker**, and **CI/CD** internal practices across multiple projects.

WATonomous

Software Engineer

July. 2025 – Present

Waterloo, ON

- Built an Automated Power Outage Notification System, leveraging **cloud-based services** and containerized microservices with Docker, reducing manual reporting delays by **80%**.
- Developed cloud-native APIs and automation scripts to integrate with SLURM and Kubernetes, enabling **parallel execution** of simulation jobs and increasing throughput by **300%**.
- Developed and containerized backend services with Docker, deploying to cloud and **HPC clusters** via Kubernetes

LandingX

CEO & Founder

July 2023 – May 2024

Remote

- Engineered modern, high-performance web applications with React.js, Node.js, Three.js, and Tailwind CSS, delivering seamless **cross-device compatibility** for **1000s of users**.
- Deployed and optimized applications using Vercel and AWS, **reducing load times by 30%** through performance tuning and **CDN utilization**.
- Managed full-stack development workflows, leveraging Vite, npm, and Git for version control and deployment.

Projects

PrepPal | SQL, Python, JS, React, Tailwind, Flask, Google Cloud, OpenCV, OpenAI, Cohere, Recharts

🔗 [repo link](#)

- Developed a **full-stack** interview preparation assistant to provide interactive mock interviews and AI-driven feedback.
- Integrated OpenAI and Cohere APIs to formulate personalized GenAI feedback and **real-time response analysis**, improving user preparation by **40%**, resulting in **100+ unique users** within 2 weeks of launch.
- Designed and implemented a structured SQL database, efficiently storing user interview history, AI-generated feedback, and progress metrics.

ASD Prediction Model | Python, Colab, NumPy, Pandas, Matplotlib, sklearn, pickle

🔗 [repo link](#)

- Achieved **93%** cross-validation accuracy in detecting Autism Spectrum Disorder using **Decision Tree** and **Random Forest** models.
- Processed data with NumPy, pandas, Matplotlib, and SMOTE to handle imbalances and extract insights.
- Optimized models in Google Colab, leveraging scikit-learn for **hyperparameter tuning** and pickle for model storage.

Speed Estimation & Vehicle Tracker | Python, OpenCV, ultralytics, NumPy, Argsparse

🔗 [repo link](#)

- Built a **CNN-based** vehicle detection and tracking model capable of calculating vehicle speed, counting vehicles, and classifying types for **detailed traffic analysis**.
- Integrated real-time video processing with Supervision & OpenCV, leveraging **tqdm to optimize performance**.
- Streamlined model deployment and inference with requests, ensuring **efficient** and accurate processing for large-scale traffic datasets and obscure camera angles.